



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

TECHNICAL PRESENTATION

CSA0303-Data Structures

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🎵 1) Music Streaming Playlist System

Dynamic Playlist using Doubly Linked List

- Manages songs efficiently
- Supports dynamic operations



Problem Statement

Need for Efficient Playlist

- Songs must be added anytime
- Songs must be removed easily
- Playlist order should be changeable
- System should be flexible



Data Structure Used

Doubly Linked List

- Each node stores one song
- Contains next and previous pointers
- Allows two-way traversal
- No fixed size limitation



Advantages

Why Doubly Linked List

- Easy insertion without shifting
- Easy deletion of songs
- Supports forward and backward play
- Reordering is simple using links
- Efficient memory usage



2) Dictionary System – Tree

Fast Word Search using Tree

- Efficient dictionary structure
- Supports quick lookup



Problem Statement

Need for Fast Search

- Large number of words
- Quick search required
- Efficient storage needed
- Avoid repeated comparisons



Data Structure Used

Tree (Prefix Tree)

- Stores words letter by letter
- Nodes represent characters
- Common prefixes are shared
- Forms a tree structure



Advantages

Why Tree

- Fast searching
- Efficient prefix matching
- Reduces redundant storage
- Suitable for autocomplete
- Works well for large data

⚙ 3) Task Scheduling – Priority Queue

- **Task Scheduling using Priority Queue**
- Manages tasks based on priority
- Ensures efficient execution



Problem Statement



Need for Scheduling

- Tasks have different priorities
- Important tasks must execute first
- System should be efficient
- Avoid delays in execution

Data Structure Used

Priority Queue (Heap)

- Stores tasks with priority
- Highest priority at top
- Uses heap structure
- Automatically sorted



Advantages



Why Priority Queue

- Executes highest priority first
- Fast insertion and deletion
- Reduces processing time
- Improves system performance
- Widely used in OS

Thankyou !